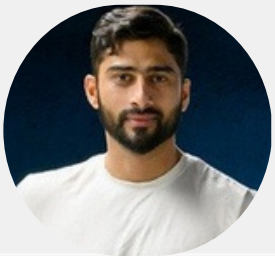
MUHAMMAD SHAHZAIB

Environmental professional with 4+ years of experience in emissions monitoring, environmental compliance, and data-driven analysis supporting reporting, audits, and operational decision-making.



mszaibch@gmail.com

+358 44 9285931

Helsinki, Finland

EDUCATION

Master's in Environmental Technology

University of Eastern Finland (2023 – 2025)

Bachelor's in Environmental Science

University of the Punjab (2016 – 2020)

SKILLS & KNOWLEDGE

Measurements and Instrumentation

- Source emission testing
- CEM systems and sensors
- Instrumentation calibration
- Sample collection and laboratory analysis

Compliance and Sustainability Frameworks

- EU IED & BAT compliance
- ISO 14001 / 9001
- GHG accounting and management
- ESG (CSRD, ESRS, ISSB, double materiality)
- Voluntary Carbon Markets (VCM) and carbon credits

Data Analysis and Visualization

- Computational analysis
- Data modeling
- Time series and statistical analysis
- Basic Machine Learning (ML)

Digital Tools and Programming

- Python, MATLAB, SQL
- Power BI, MS Fabric, Looker, Tableau
- Digital (MRV) dashboards
- Descision support analytics
- Cloud and Data platforms

CERTIFICATIONS

- ESG (The ESG Institute)
- Carbon Dioxide Removal - CDR (NASDAQ)
- Data Analytics (Deloitte)
- Quantitative Research (JP Morgan)
- Data Engineering DP-600 (Microsoft Fabric)
- Occupational Safety (Kiwa Suomi)
- Embodied Carbon, LCA, EPD (Paroc Oy)
- Project management (Talent Hub Eastern Finland)
- Professional competence certificate (YUFE)

LANGUAGES

English (*Advanced*)
Finnish (Satisfactory)

PORTFOLIO

https://ms-muhammad-shahzaib-portfolio-webs.vercel.app/

PROFESSIONAL EXPERIENCE

Environmental Associate | Environmental Engineering

Paroc, Finland | 2025- Present

Manufacturer of stone wool insulation solutions for energy-efficient and sustainable construction

- Executed industrial emission measurement campaigns in a stone wool manufacturing facility in line with EN ISO standards, EU IED, and EU ETS requirements.
- Developed data processing and visualization workflows using Python, Excel, and Power BI to improve emissions data traceability and decision-making.
- Integrated P&ID diagrams, process flow schematics, and time-series process/emissions data to analyze emission behavior and process performance.
- Applied correlation and regression analysis to identify emission trends and evaluate abatement systems, including cyclone separators, bag filters, and droplet separators.
- Identified and resolved discrepancies between in-house monitoring systems and accredited third-party measurements during compliance activities with Ramboll and DGE.
- Delivered technical recommendations for process optimization, emissions monitoring improvements, and compliance follow-up actions to management.

Research Assistant | Environmental & Emissions Research

University of Eastern Finland | 2023 - 2024

FINE Aerosol Research Group (ILMARI Lab) – aerosol and emissions research

- Contributed to EU-funded environmental and emissions measurement projects, supporting laboratory experiments, monitoring campaigns, and technical reporting.
- Processed, quality-controlled, and analyzed environmental datasets from laboratory and field measurements to evaluate emissions trends and air quality impacts.
- Supported setup, calibration, and troubleshooting of emissions sampling and measurement systems to ensure data reliability and experimental accuracy.
- Developed a small environmental modeling project using SMEAR II monitoring station data to assess seasonal and annual emission trends in the Helsinki region.
- Co-authored scientific publications, conference papers, and technical reports in collaboration with international research partners.

Monitoring and Evaluation Officer

City District Government Lahore | 2021- 2022

Planning and Development Department

- Supported monitoring and evaluation of urban development initiatives in a fast-paced municipal environment, contributing to service delivery and operational performance across a major city of around 11 million residents.
- Coordinated with field and administrative teams to support data collection, verification, and operational alignment across urban service programs, while assisting with reporting and tracking using Excel and GIS/GPS tools.
- Prepared structured performance reports and data summaries for senior municipal stakeholders to support decision-making in critical urban infrastructure and service operations.

Environmental Coordinator

Integrated Consulting Services Limited | 2020- 2021

Environmental consulting and compliance services

- Conducted EMS and QMS audits, ensuring environmental and regulatory compliance; supported onsite data collection and EIA reporting
- Prepared carbon accounting inputs, waste assessments, lab analysis, and compliance documentation for sustainability reporting and certifications